

Biography

I am a Postdoctoral Researcher in Robotics at Mohamed bin Zayed University of Artificial Intelligence (MBZUAI). My research focuses on embodied AI, SLAM, autonomous navigation, and agricultural robotics. During my Ph.D., I developed active SLAM systems that integrate GPS, digital maps, and road segmentation for large-scale outdoor mapping, localization, and exploration. My current research investigates embodied AI frameworks for autonomous fruit harvesting through tactile-vision-action models, contact-rich simulation of deformable fruits and stems, synthetic data generation, and robotic manipulation.

I have authored more than 10 publications in leading robotics conferences and journals, including IROS, ICRA, and RAL, accumulating over 460 citations and an h-index of 8. I actively serve as a reviewer for major robotics conferences and journals, including ICRA, IROS, CASE, and RAL and as Co-Chair of a Special Session at the 2027 IEEE Symposium Series on Computational Intelligence (SSCI). Beyond academia, I co-founded Fruitbot, an agricultural robotics startup translating embodied AI research into autonomous harvesting systems, and have extensive hands-on experience deploying wheeled, quadruped, and manipulator robots in real-world applications involving SLAM, navigation, manipulation, reinforcement learning, and autonomous operation.

Education

Mohamed bin Zayed University of Artificial Intelligence

POSTDOC IN ROBOTICS

Abu Dhabi, UAE

Jul. 2024 - Present

Nanjing University of Science and Technology

PH.D. IN COMPUTER SCIENCE

Nanjing, China

Sep. 2017 - Jun. 2024

Tongji University

B.S. IN ELECTRONIC AND INFORMATION ENGINEERING

Shanghai, China

Sep. 2013 - Jun. 2017

Professional Experience

Fruitbot Limited

ABU DHABI, UAE

Founder & CTO

Jul. 2025 - Present

- Co-founded Fruitbot, an agricultural robotics startup developing autonomous fruit-harvesting systems powered by embodied AI and intelligent robotic manipulation.
- Led the design of an end-to-end embodied AI framework for autonomous fruit harvesting, encompassing tactile-vision-action foundation models, physics-based simulation of contact-rich interactions with deformable fruits and stems for synthetic data generation and policy training, and task-specific strawberry gripper hardware for reliable perception and manipulation.
- Secured AED 70,000 in competitive seed funding through the MBZUAI Entrepreneurship and Incubation Program.
- Developed an autonomous robotic harvesting prototype that was showcased and demonstrated to H.H. Sheikh Mohamed bin Zayed Al Nahyan, President of the United Arab Emirates.

NiceCart

WUXI, CHINA

Robotics System Technical Expert

Aug. 2026 - Present

- Leading the lab-to-field delivery of university-developed agricultural robotics technologies, translating research prototypes into deployable autonomous harvesting systems for real-world commercial greenhouse operations.
- Integrating, deploying, and validating the complete robotic harvesting stack in real farms, spanning autonomous navigation, perception, motion planning, and manipulation, while bridging research development with practical agricultural requirements and operational constraints.

Horizon Robotics

RIYADH, SAUDI ARABIA

Invited Robotics Trainer

Apr. 2026 - May. 2026

- Delivered professional training on the Unitree Go2 quadruped robot, Unitree B2-W wheeled-legged robot, and Unitree Z1 robotic arm.
- Provided hands-on workshops on SLAM, autonomous navigation, robot manipulation, perception, and reinforcement learning.
- Trained engineers and researchers in Isaac Sim-based reinforcement learning, sim-to-real policy transfer, and deployment of autonomous locomotion and manipulation policies on real-world robotic platforms.

Alpha-Z AI Limited

IRVINE, USA

Founding Member

Apr. 2024 - Nov. 2024

- Founded an AI and robotics startup focused on embodied intelligence and autonomous robotic systems.
- Led the architecture and deployment of end-to-end autonomy solutions, integrating multi-sensor fusion, SLAM, localization, navigation, and decision-making for real-world robotic platforms.

University of Macau

MACAU, CHINA

Research Assistant

Mar. 2022 – Mar. 2024

- Investigated large-scale outdoor active mapping and autonomous exploration in unknown environments.
- Developed active SLAM algorithms integrating GPS, digital maps, and road segmentation for efficient mapping, localization, and navigation in large-scale outdoor environments.

Zhejiang Lab

HANGZHOU, CHINA

Research Intern

May 2020 – Aug. 2020

- Developed indoor 2D mapping and autonomous exploration algorithms for mobile robots operating in unknown environments.

Horizon Robotics

NANJING, CHINA

Research Intern

Apr. 2019 – Jan. 2020

- Developed LiDAR-based global localization algorithms for autonomous robotic systems.

Publications

PEER-REVIEWED PUBLICATIONS

- Haoran Ding, Anqing Duan, **Ze Zhou Sun**, Dezhen Song, and Yoshihiko Nakamura, “Virtual In-Hand View Synthesis for Robust Visuomotor Policies: A Case Study on Strawberry Picking” *IROS 2026*
- Haoran Ding, Anqing Duan, **Ze Zhou Sun**, Leonel Roza, Noémie Jaquier, Dezhen Song, and Yoshihiko Nakamura, “Towards Safe Imitation Learning via Potential Field-Guided Flow Matching” *IROS 2025*
- Wei Gao, **Ze Zhou Sun**, Mingle Zhao, Cheng-Zhong Xu, and Hui Kong, “Active Loop Closure for OSM-Guided Robotic Mapping in Large-Scale Urban Environments” *IROS 2024*
- Ze Zhou Sun**, Banghe Wu, Chengzhong Xu, and Hui Kong, “Concave-Hull Induced Graph-Gain for Fast and Robust Robotic Exploration” *RAL 2023*
- Ze Zhou Sun**, Banghe Wu, Chengzhong Xu, and Hui Kong, “Ada-Detector: Adaptive Frontier Detector for Rapid Exploration” *ICRA 2022*
- Yigong Zhang, **Ze Zhou Sun**, Banghe Wu, Jian Yang, and Hui Kong, “Building a Stereo and Wide-View Hybrid RGB/FIR Imaging System for Autonomous Vehicle” *IEEE Sensors Journal 2021*
- Ying Wang, **Ze Zhou Sun**, Cheng-Zhong Xu, Sanjay E. Sarma, Jian Yang, and Hui Kong, “Lidar Iris for Loop-Closure Detection” *IROS 2020*
- Ze Zhou Sun**, Banghe Wu, Cheng-Zhong Xu, Sanjay E. Sarma, Jian Yang, and Hui Kong, “Frontier Detection and Reachability Analysis for Efficient 2D Graph-SLAM Based Active Exploration” *IROS 2020*
- Minghao Liu, Zhixing Hou, **Ze Zhou Sun**, Ning Yin, Hang Yang, Ying Wang, Zhiqiang Chu, and Hui Kong, “Campus Guide: A LiDAR-Based Mobile Robot” *ECMR 2019*
- Yigong Zhang, Yicheng Gao, Shuo Gu, Yubin Guo, Minghao Liu, **Ze Zhou Sun**, Zhixing Hou, Hang Yang, Ying Wang, Jian Yang, et al., “Build Your Own Hybrid Thermal/EO Camera for Autonomous Vehicle” *ICRA 2019*